

VISTA

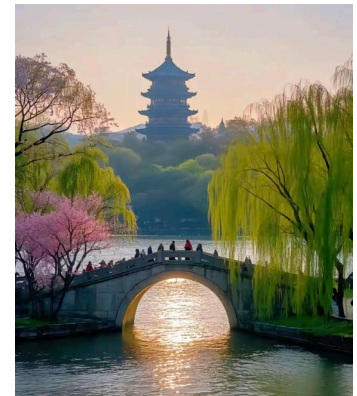
Virtual International Seminar on Theoretical Advancements
Seminar 112



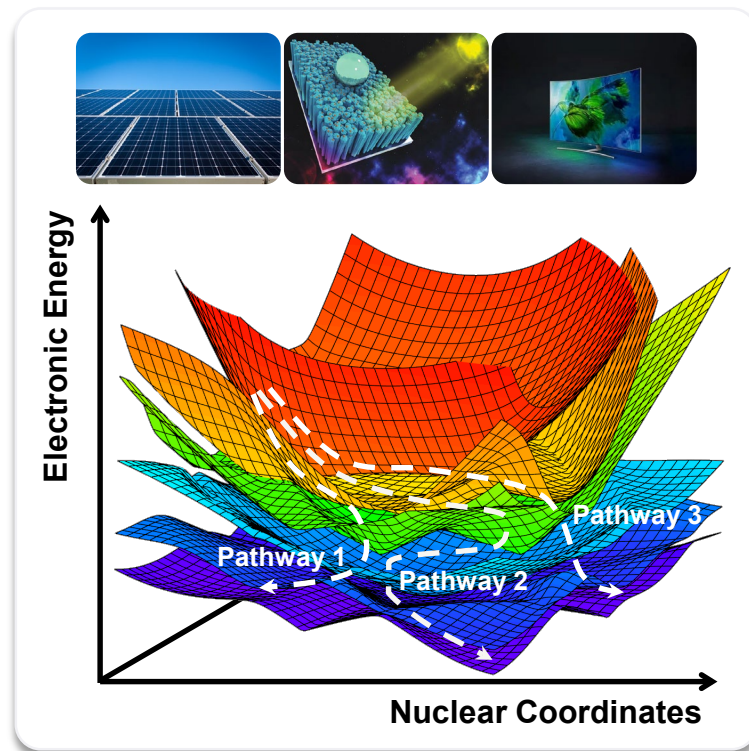
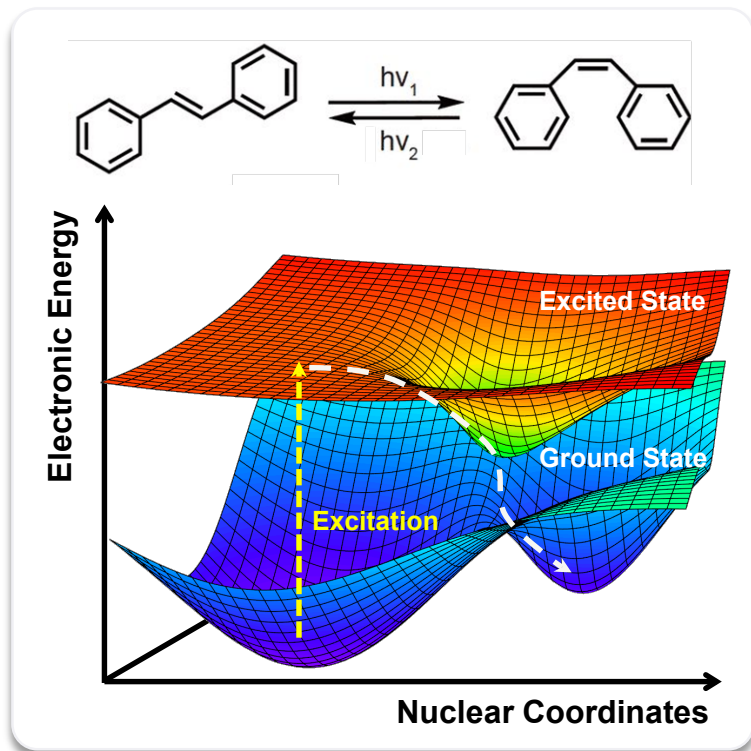
Detailed Complementary Consistency for Mixed Quantum-Classical Nonadiabatic Dynamics



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Zhejiang University, China
September 16, 2026



Nonadiabatic Dynamics



➤ **Nonadiabatic dynamics methods (not a complete list):** $\frac{\partial \psi(\{\vec{R}_\alpha\}, \{\vec{r}_i\}, t)}{\partial t} = \frac{1}{i\hbar} \hat{H} \psi(\{\vec{R}_\alpha\}, \{\vec{r}_i\}, t)$

DVR, MCTDH, DMRG, HEOM, DEOM, SSE, AIMS, MQCL, QTMF, CTMQC, EMF, TSH, SCDM, MM, NAF, MASH, DCC ...

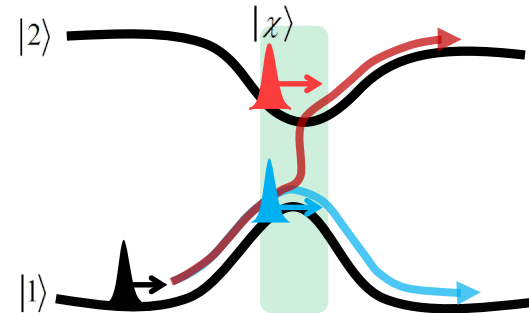
Quantum dynamics has intrinsic “curse of dimensionality”
Different methods have different advantages and limitations

Quantum Description of Nonadiabatic Dynamics

Direct product state $|\Psi\rangle = |\chi\rangle|\psi\rangle = |\chi\rangle(w_1|1\rangle + w_2|2\rangle)$

$$i\hbar \frac{\partial |\Psi\rangle}{\partial t} = (\hat{T}_n + \hat{H}_{el})|\Psi\rangle \xrightarrow{\text{Classical limit}} i\hbar \frac{\partial |\psi\rangle}{\partial t} = \hat{H}_{el}|\psi\rangle$$

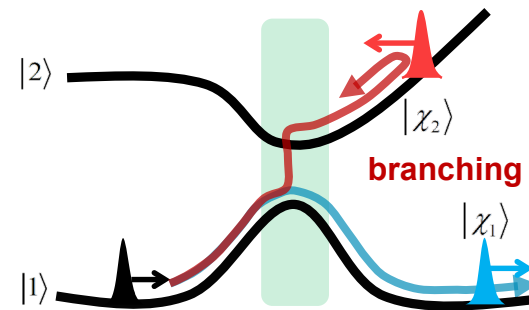
Time-dependent propagation $\dot{\sigma} = \frac{1}{i\hbar} \left[\hat{H} - i\hbar \sum_{\alpha} \mathbf{v}_{\alpha} \cdot \mathbf{d}^{\alpha}, \sigma \right]$



Entangled state $|\Psi\rangle = |\chi_1\rangle|1\rangle + |\chi_2\rangle|2\rangle$

$\rho = |\Psi\rangle\langle\Psi| \xrightarrow{\text{decay with time}} \sigma = \text{Tr}_n[\rho] = \begin{pmatrix} \langle\chi_1|\chi_1\rangle & \langle\chi_2|\chi_1\rangle \\ \langle\chi_1|\chi_2\rangle & \langle\chi_2|\chi_2\rangle \end{pmatrix}$

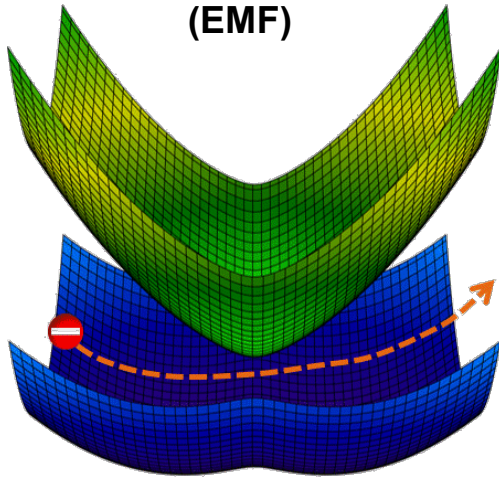
Quantum decoherence $\dot{\sigma} = \frac{1}{i\hbar} \left[\hat{H} - i\hbar \sum_{\alpha} \mathbf{v}_{\alpha} \cdot \mathbf{d}^{\alpha}, \sigma \right] - \mathcal{R}[\sigma]$



Branching and decoherence are essential for nonadiabatic dynamics

Mixed Quantum-Classical Dynamics (MQCD)

Ehrenfest Mean Field (EMF)



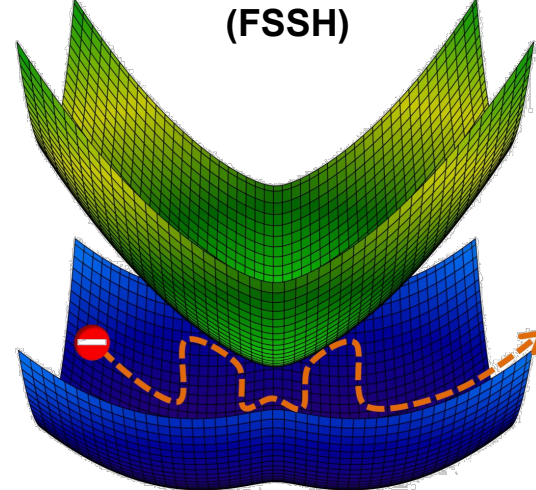
Electron evolution $i\hbar\partial_t|\psi\rangle = \hat{H}_e|\psi\rangle$

Nuclear dynamics $m_\alpha\ddot{\mathbf{x}}_\alpha = -\nabla_\alpha\langle\psi|\hat{H}_e|\psi\rangle$

Average surface $E_{eff} = \sum_i |w_i|^2 E_i$

Ehrenfest, *Z. Physik* 45, 455 (1927)

Fewest Switches Surface Hopping (FSSH)



Electron evolution $i\hbar\partial_t|\psi\rangle = \hat{H}_e|\psi\rangle$

Nuclear dynamics $m_\alpha\ddot{\mathbf{x}}_\alpha = -\nabla_\alpha\langle\phi_\lambda|\hat{H}_e|\phi_\lambda\rangle$

Hopping probabilities $g_{\lambda i} = \max(0, -b_{\lambda i}\Delta t / \sigma_{\lambda\lambda})$

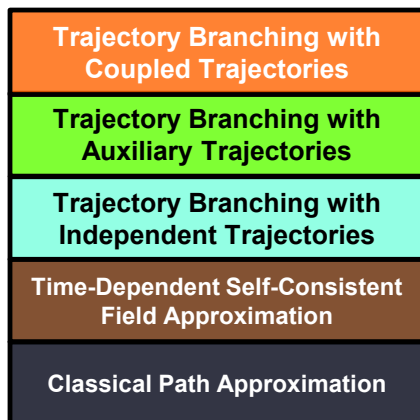
Tully, *J. Chem. Phys.* 93, 1061 (1990)

EMF trajectory is deterministic, while FSSH ensemble has branching
More robust theoretical framework for MQCD is still needed

Main Research Interests

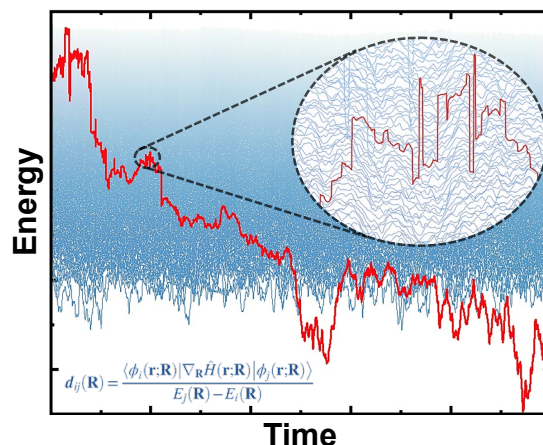
Consistent Theory

Electron-Nuclei Correlation ↑



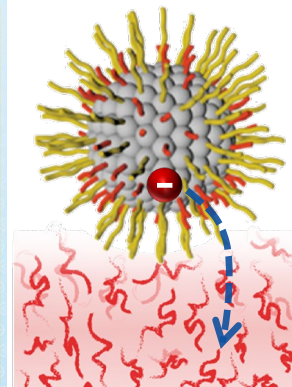
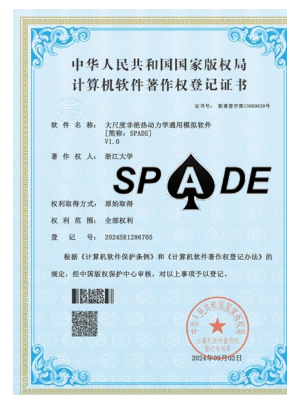
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Large-Scale Methods



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Dynamics Mechanisms



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Detailed Complementary Consistency for MQCD

- **Statistical Interpretation of MQCD**
- Detailed Complementary Consistency
- Some Related Method Developments

Quantum Dynamics



Statistical interpretation of quantum mechanics

The quantity $|\Psi|^2$ has an important physical interpretation: It is related to the probability that a system can be found in a particular region of space

	Quantum dynamics		MQCD
State description	$ \psi\rangle$	$\hat{\rho} = \psi\rangle\langle\psi $	?
Statistical interpretation	$P_i = \langle i \psi\rangle ^2$	$P_i = \rho_{ii}$	?
Expectation value of an observable	$\langle\hat{O}\rangle = \langle\psi \hat{O} \psi\rangle$	$\langle\hat{O}\rangle = \text{Tr}(\hat{\rho}\hat{O})$	?
Equation of motion	$\frac{\partial \psi\rangle}{\partial t} = \frac{1}{i\hbar}\hat{H} \psi\rangle$	$\frac{\partial\hat{\rho}}{\partial t} = \frac{1}{i\hbar}[\hat{H}, \hat{\rho}]$	?

Born, *Science* 122, 675 (1955); *Physics in My Generation*, Springer (1969)

Mixed Quantum-Classical Liouville (MQCL)

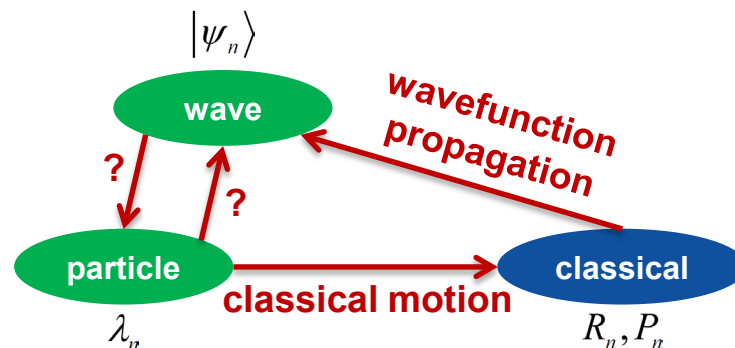
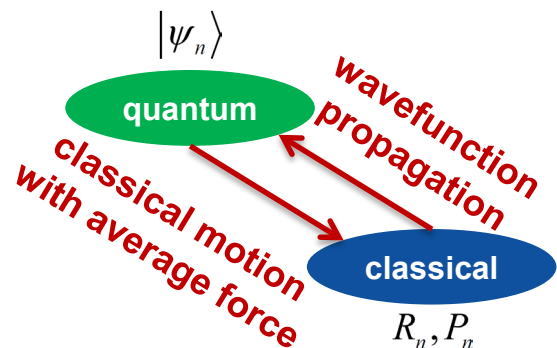
	MQCL
State description	$\hat{\rho}^W(R, P)$
Statistical interpretation	$P_i = \int \int \hat{\rho}_{ii}^W(R, P) dR dP$
Expectation value of an observable	$\langle \hat{O} \rangle = Tr \left(\int \int \hat{\rho}^W \hat{O}^W dR dP \right)$
Equation of motion	$\frac{\partial \hat{\rho}^W(R, P)}{\partial t} = \frac{1}{i\hbar} [\hat{H}^W, \hat{\rho}^W] + \frac{1}{2} \left(\{ \hat{H}^W, \hat{\rho}^W \} - \{ \hat{\rho}^W, \hat{H}^W \} \right)$ $\frac{\partial \rho_{ij}^W(R, P)}{\partial t} = \frac{1}{i\hbar} (E_i(R) - E_j(R)) \rho_{ij}^W - \frac{P}{M} \cdot \nabla_R \rho_{ij}^W$ $- \frac{1}{2} \sum_k \left(F_{ik}^W \cdot \nabla_P \rho_{kj}^W + \nabla_P \rho_{ik}^W \cdot F_{kj}^W \right) + \sum_k \frac{P}{M} \cdot \left(\rho_{ik}^W d_{kj} - d_{ik} \rho_{kj}^W \right)$

It is still challenging to solve the MQCL equation efficiently

Kapral and Ciccotti, *J. Chem. Phys.* 110, 8919 (1999)

Subotnik, Ouyang, and Landry, *J. Chem. Phys.* 139, 214107 (2013)

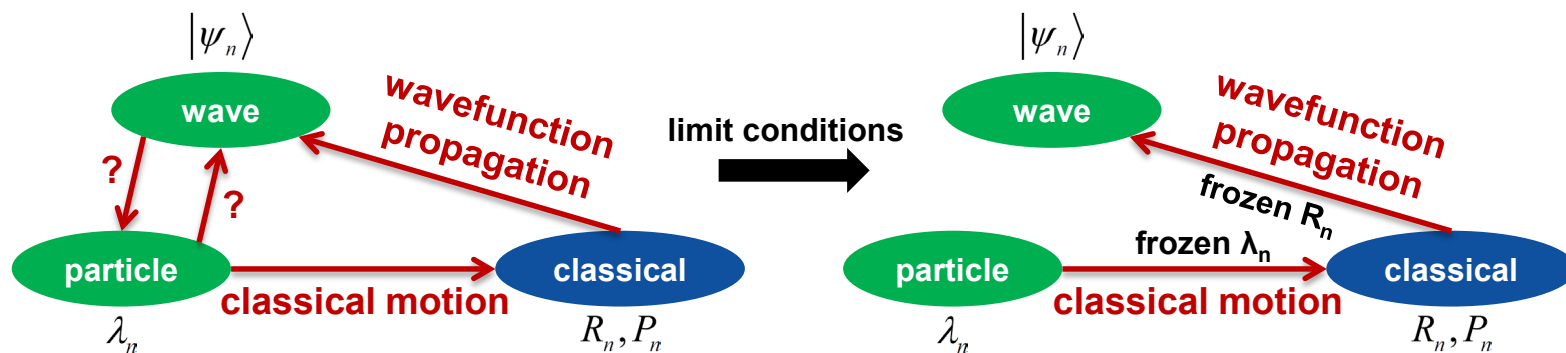
Statistical Interpretation of MQCD



	MQCD
State description	$\{ \psi_n\rangle, R_n, P_n, \lambda_n\}$
Statistical Interpretation	$\rho_{ii}^{\text{wf}}(R, t) = \frac{1}{N_{\text{traj}}} \sum_{n=1}^{N_{\text{traj}}} \left \langle i \psi_n(t) \rangle \right ^2 \delta(R - R_n(t))$ $\rho_{ij}^{\text{wf}}(R, t) = \frac{1}{N_{\text{traj}}} \sum_{n=1}^{N_{\text{traj}}} \langle i \psi_n(t) \rangle \langle \psi_n(t) j \rangle \delta(R - R_n(t))$ $\rho_{ii}^{\text{as}}(R, t) = \frac{1}{N_{\text{traj}}} \sum_{n=1}^{N_{\text{traj}}} \delta_{i, \lambda_n(t)} \delta(R - R_n(t))$

L. Wang* et al., *J. Phys. Chem. Lett.* 15, 6771 (2024)

Statistical Interpretation of MQCD



	MQCD
Expectation value of an observable	$\langle \hat{O} \rangle = Tr \left(\int \hat{\rho}^{\text{wf}} \hat{O} dR \right) \quad \langle \hat{O} \rangle = \sum_i \int \rho_{ii}^{\text{as}} O_{ii} dR$ $\langle \hat{O} \rangle^{\text{wf}} = \frac{1}{N_{\text{traj}}} \sum_n^{N_{\text{traj}}} Tr \left(\hat{\rho}_n^{\text{wf}} \hat{O}(R_n) \right) \quad \langle \hat{O} \rangle^{\text{as}} = \frac{1}{N_{\text{traj}}} \sum_n^{N_{\text{traj}}} O_{\lambda_n \lambda_n}(R_n)$
Equation of motion	$\frac{\partial \psi_n\rangle}{\partial t} = \frac{1}{i\hbar} \hat{H}_e(R_n) \psi\rangle + \mathcal{X} \quad \lambda_n(t) \xrightarrow{z} \lambda_n(t+dt)$ $\frac{dP_n}{dt} = -\nabla_R E_\lambda(R_n) + \mathcal{Y} \quad \frac{dR_n}{dt} = \frac{1}{M} P_n$

L. Wang* et al., *J. Phys. Chem. Lett.* 15, 6771 (2024)

Principle of Detailed Internal Consistency

Basic requirement of consistency at the ensemble level: $\rho_{ii}^{\text{wf}}(R, t) = \rho_{ii}^{\text{as}}(R, t)$

	MQCD
State description	$\{ \psi_n\rangle, R_n, P_n, \lambda_n\}$
Statistical interpretation	$\rho_{ij}(R, t) = \frac{1}{N_{\text{traj}}} \sum_{n=1}^{N_{\text{traj}}} \langle i \psi_n(t) \rangle \langle \psi_n(t) j \rangle \delta(R - R_n(t))$
Expectation value of an observable	$\langle \hat{O} \rangle = \frac{1}{N_{\text{traj}}} \sum_n Tr(\psi_n\rangle \langle \psi_n \hat{O}(R_n))$
Equation of motion	$\frac{\partial \psi_n\rangle}{\partial t} = \frac{1}{i\hbar} \hat{H}_e(R_n) \psi_n\rangle + \mathcal{X} \quad \lambda_n(t) \xrightarrow{z} \lambda_n(t + dt)$ $\frac{dP_n}{dt} = -\nabla_R E_\lambda(R_n) + \mathcal{Y} \quad \frac{dR_n}{dt} = \frac{1}{M} P_n$

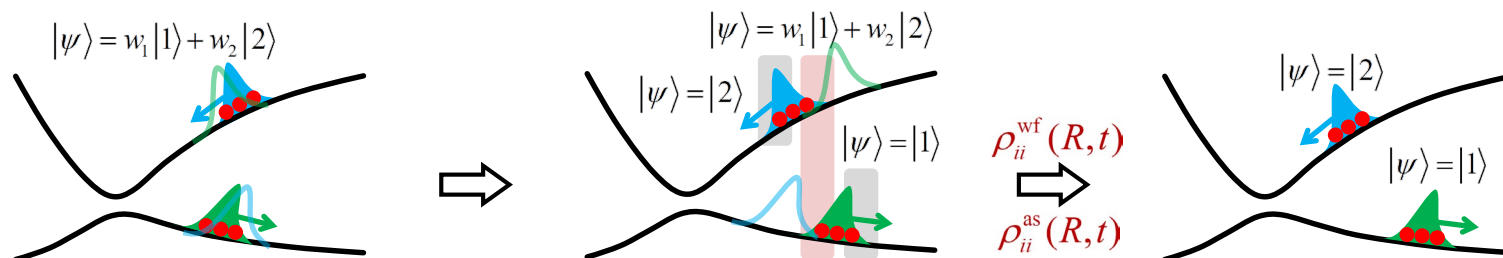
**The wave and particle descriptions
should yield identical results at any place and any time**

L. Wang* et al., *J. Phys. Chem. Lett.* 15, 6771 (2024)

Outline

- Statistical Interpretation of MQCD
- **Detailed Complementary Consistency**
- Some Related Method Developments

Detailed Complementary Consistency (DCC) Method



The population distributions are difficult to calculate; There are many possible solutions

1. Local trajectory density calculation:

$$N_i^n = \sum_{l=1}^N \Theta(\varepsilon^2 - \|\mathbf{x}_n - \mathbf{x}_l\|^2) \Theta\left(\frac{\mathbf{p}_n \cdot \mathbf{p}_l}{\|\mathbf{p}_n\| \cdot \|\mathbf{p}_l\|} - \cos(\theta)\right) \delta_{i,a_l}$$

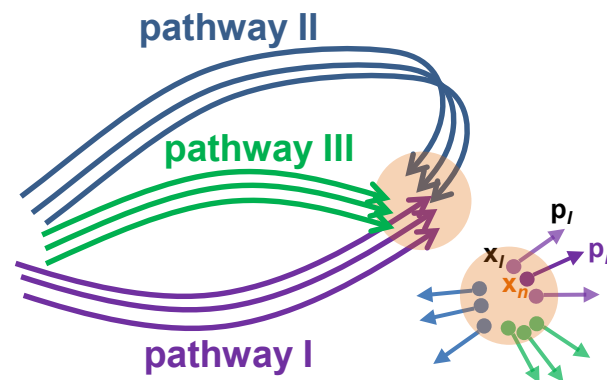
$$\Theta(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases} \quad \rho_{ii}^{n,as} = \frac{N_i^n}{\sum_j N_j^n} \quad z_n = \frac{|\rho_{11}^{n,as} - \rho_{11}^{n,wf}|}{\sqrt{\rho_{11}^{n,wf}(1 - \rho_{11}^{n,wf})} / \sqrt{\sum_i N_i^n}}$$

2. Hypothesis testing: z-test, χ^2 -test

3. Consistent correction:

$$\rho'_{ii} = \rho_{ii}^{as} \quad \rho'_{ij} = \frac{\sqrt{\rho'_{ii} \rho'_{jj}}}{|\rho_{ij}|} \rho_{ij}$$

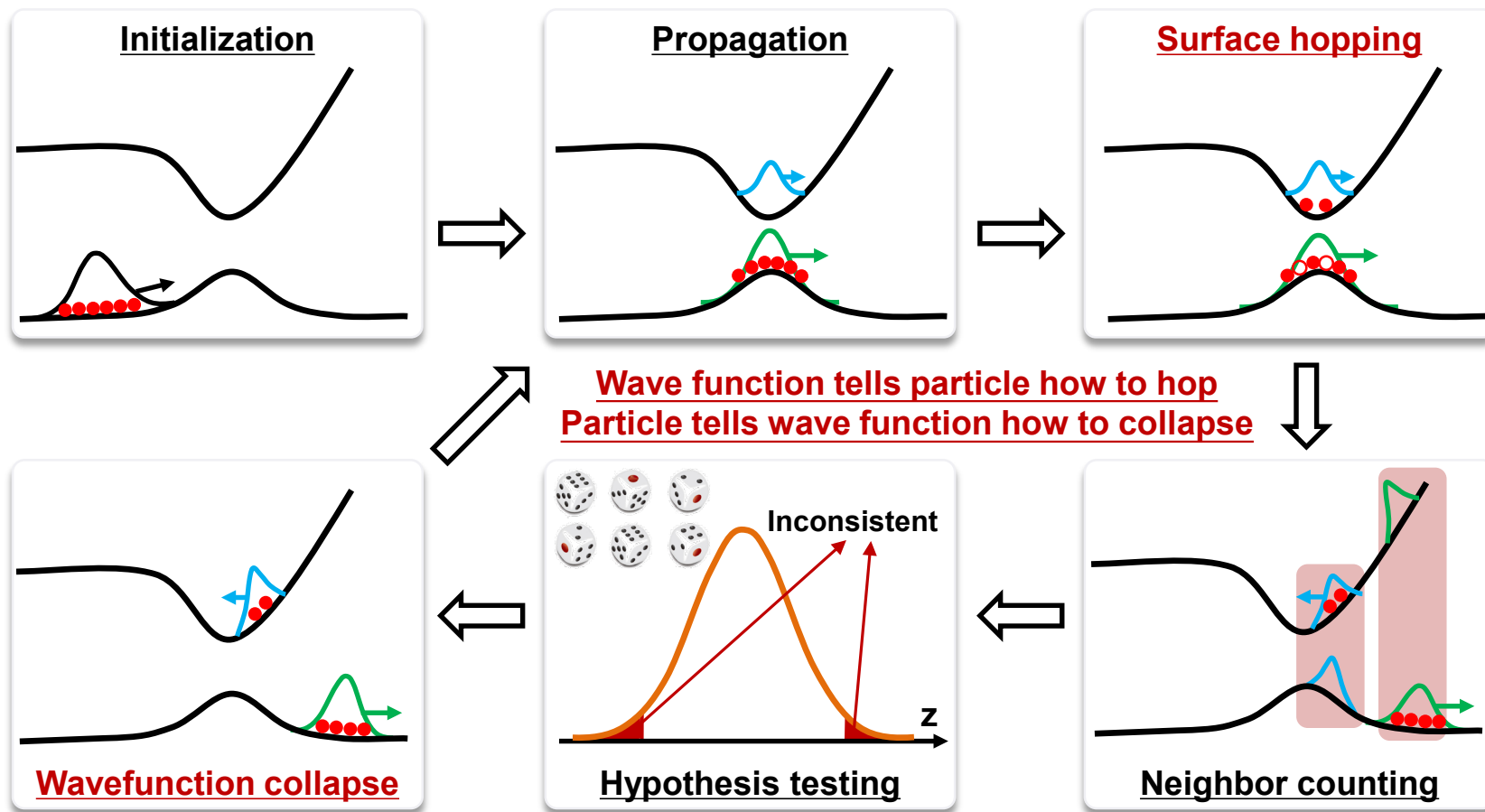
consistency is associated with pathway



If each trajectory is consistent, the trajectory ensemble becomes consistent

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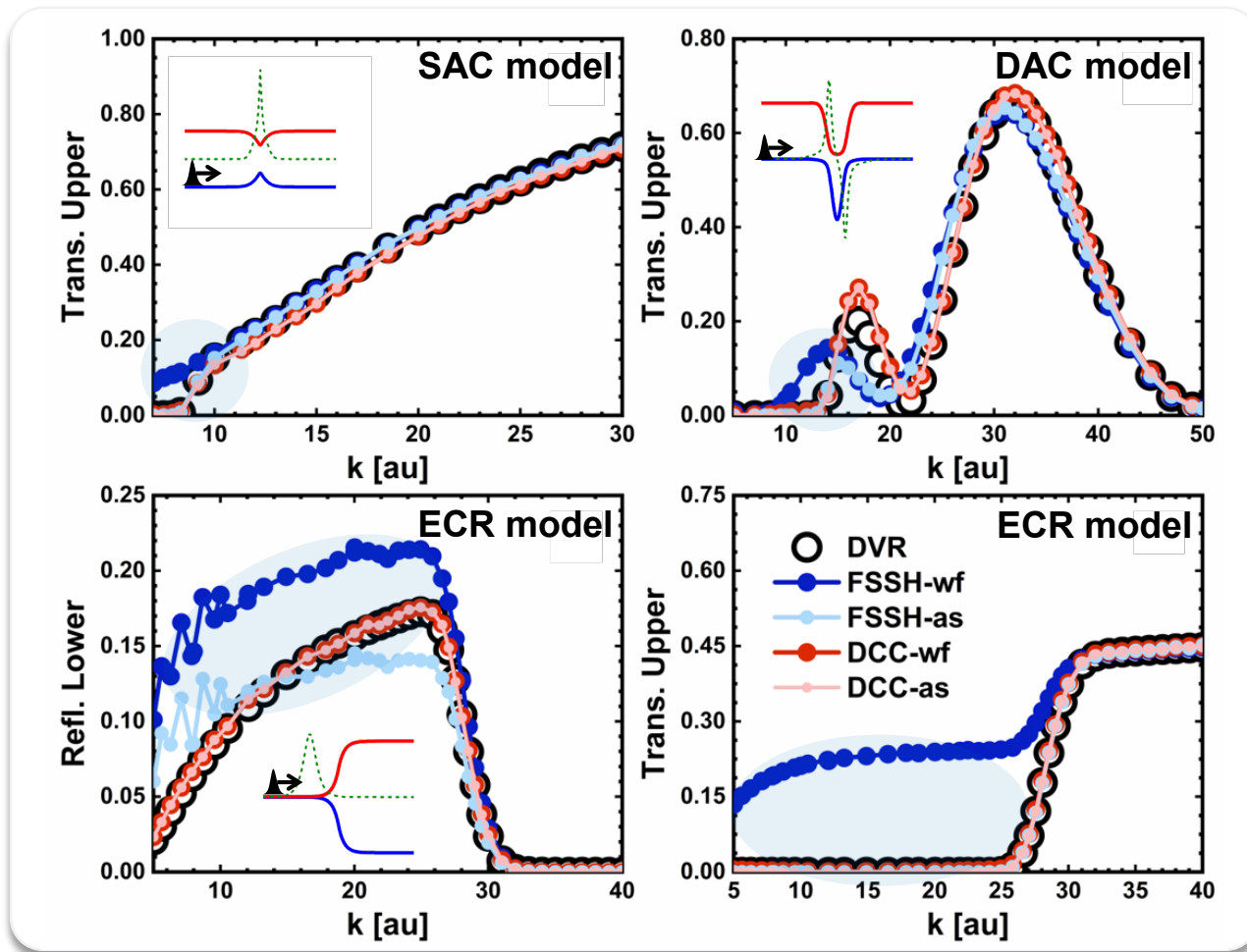
Step-by-Step Algorithms of DCC



DCC naturally gives surface hopping and trajectory branching
DCC also properly describes coherence and decoherence

L. Wang* et al., *J. Phys. Chem. Lett.* 15, 6771 (2024)

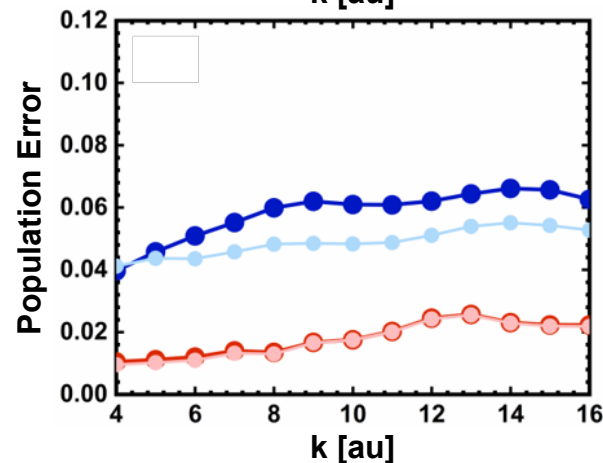
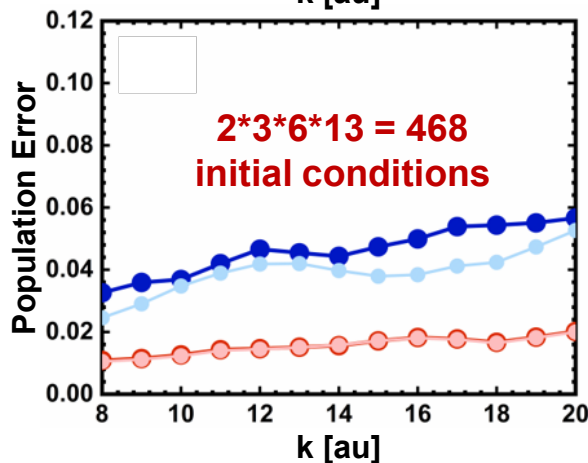
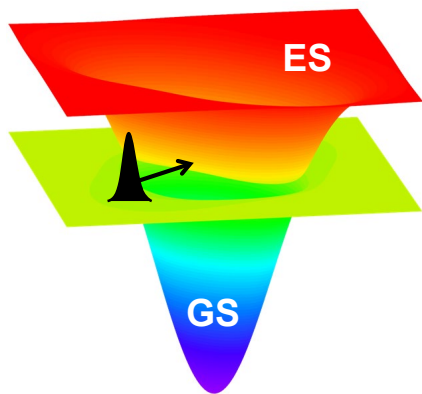
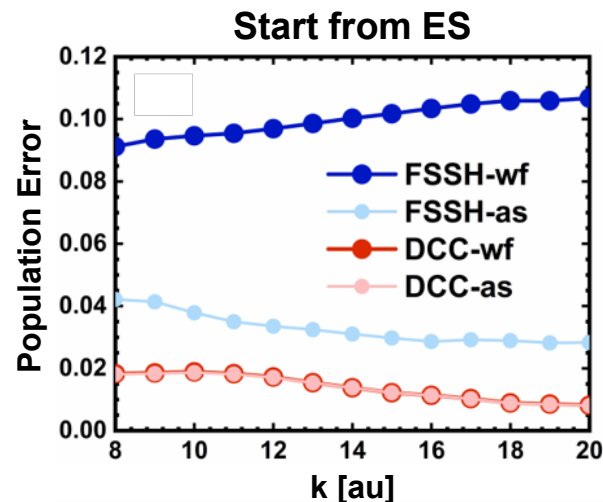
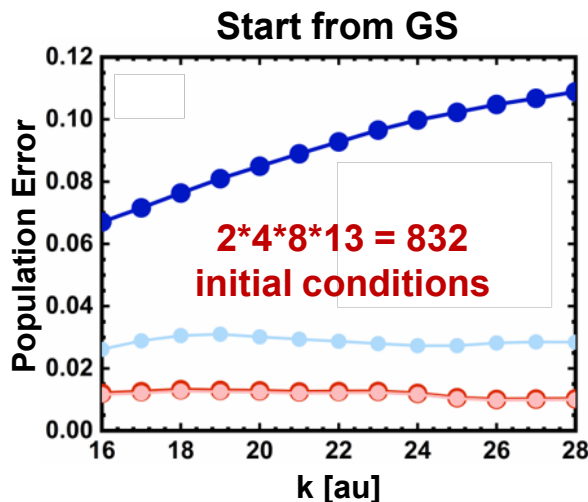
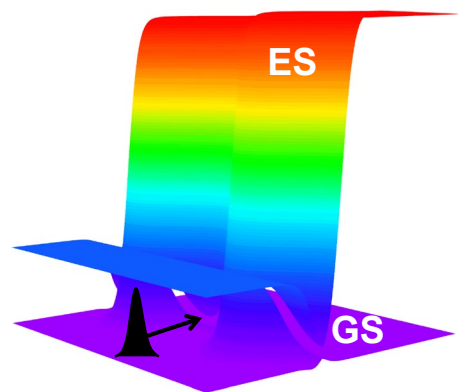
One-Dimensional Tully Scattering Models



Active states and wave functions give identical and accurate results

Tully, *J. Chem. Phys.* 93, 1061 (1990)

Two-Dimensional Representative Scattering Models

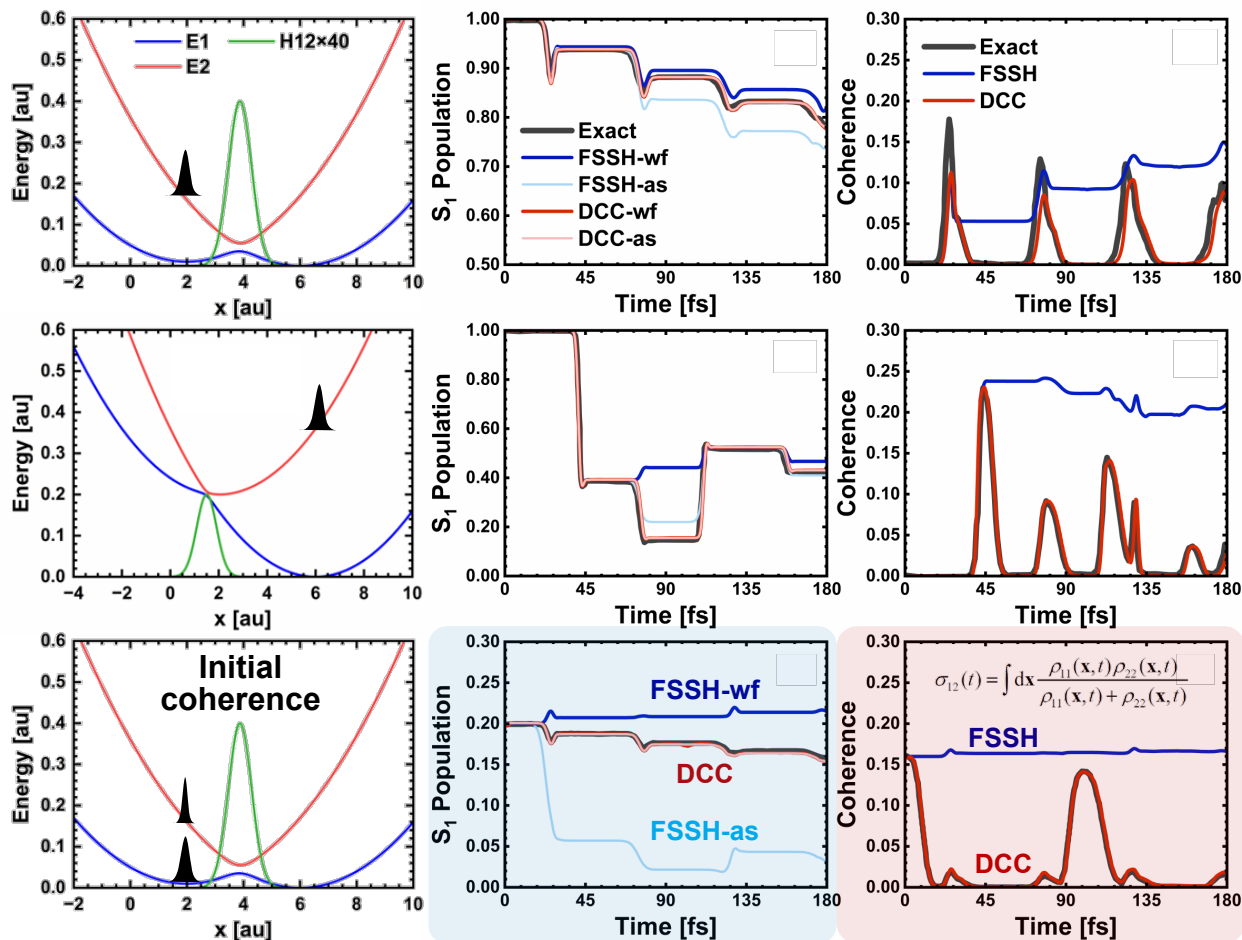


Active states and wave functions give identical and accurate results

Shenvi, Subotnik et al., *J. Chem. Phys.* 135, 024101 (2011); *J. Phys. Chem. A* 115, 12083 (2011)

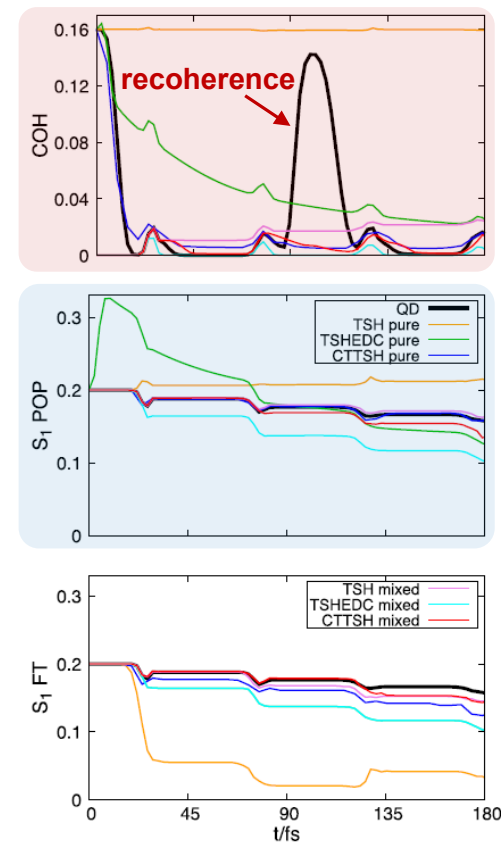
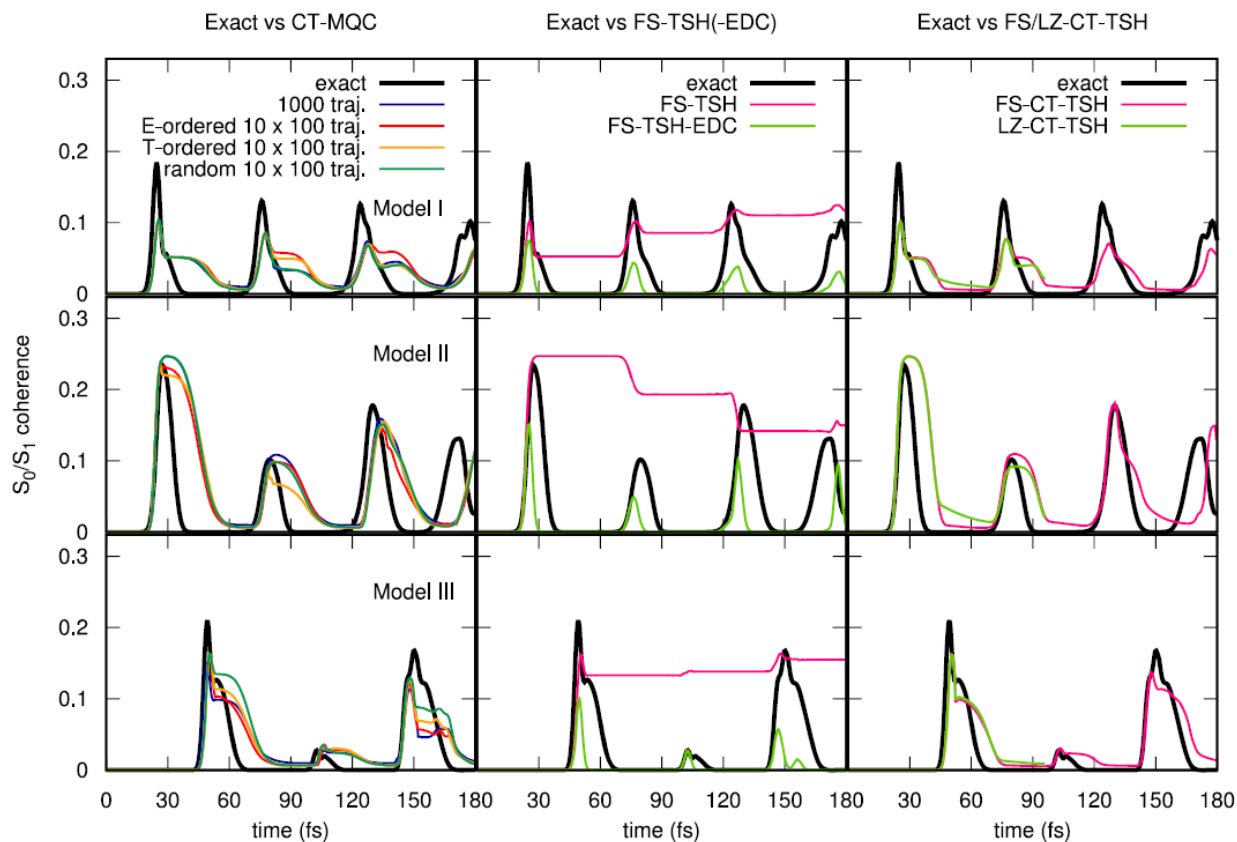
One-Dimensional Bound-State Models

Hamiltonian $H_{11}(x) = \frac{1}{2}k(R - R_1)^2$ $H_{22}(x) = \frac{1}{2}k(R - R_2)^2 + \Delta$ $H_{12}(x) = H_{21}(x) = be^{-a(R - R_3)^2}$



Agostini et al., *J. Chem. Theory Comput.* 17, 5969 (2021); *J. Chem. Phys.* 160, 054102 (2024)

One-Dimensional Bound-State Models

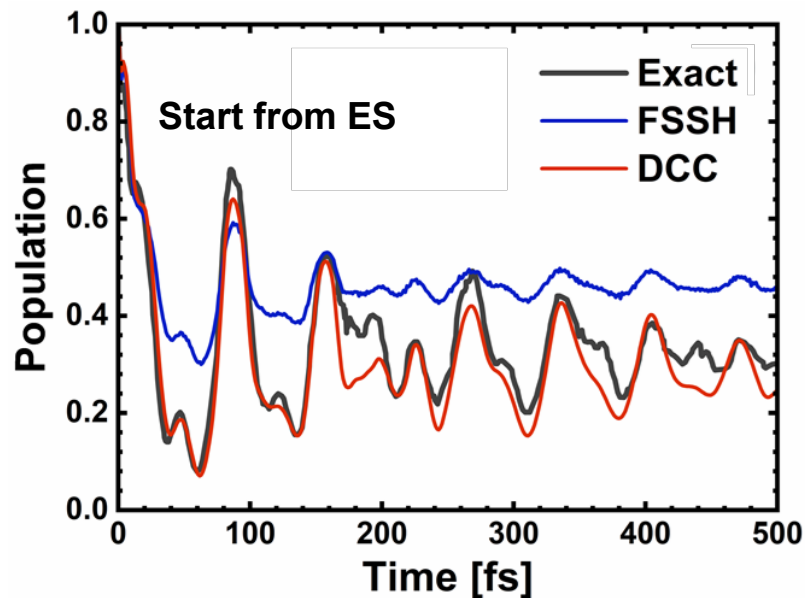
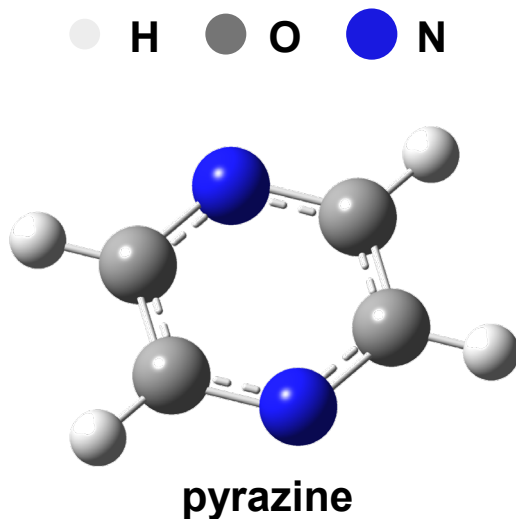


Coherence and recoherence are much more difficult to describe
DCC outperforms all other investigated methods in these models

Agostini et al., *J. Chem. Theory Comput.* 17, 5969 (2021); *J. Chem. Phys.* 160, 054102 (2024)

Three-Mode Pyrazine Model

Hamiltonian $\hat{H} = \begin{bmatrix} E_1 & 0 \\ 0 & E_2 \end{bmatrix} + \sum_{i=10a,6a,1} \frac{\omega_i}{2} (\hat{P}_i^2 + \hat{R}_i^2) + \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_{\text{nonlocal e-ph coupling}} + \underbrace{\begin{bmatrix} 0 & \lambda \\ \lambda & 0 \end{bmatrix}}_{\text{nonlocal e-ph coupling}} \hat{R}_{10a} + \underbrace{\begin{bmatrix} \kappa_{6a}^{(1)} & 0 \\ 0 & \kappa_{6a}^{(2)} \end{bmatrix}}_{\text{local e-ph couplings}} \hat{R}_{6a} + \underbrace{\begin{bmatrix} \kappa_1^{(1)} & 0 \\ 0 & \kappa_1^{(2)} \end{bmatrix}}_{\text{local e-ph couplings}} \hat{R}_1$



DCC gives correct long-time dynamics and equilibrium populations

Schneider and Domcke, *Chem. Phys. Lett.* 150, 235 (1988)

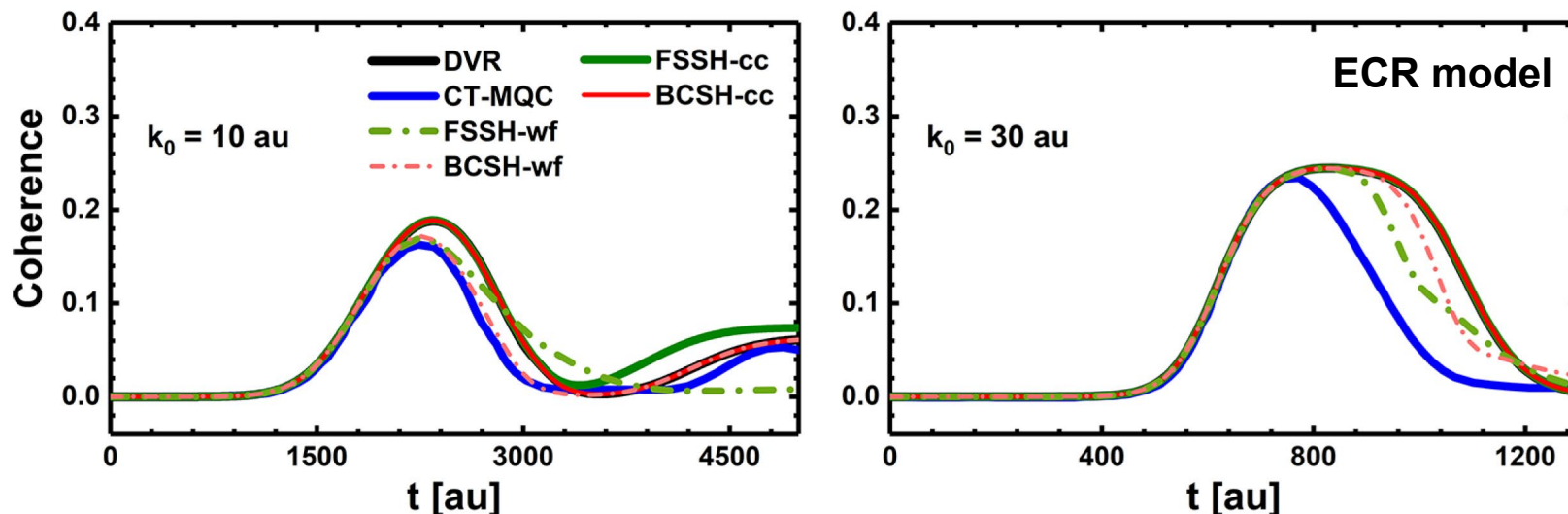
Outline

- Statistical Interpretation of MQCD
- Detailed Complementary Consistency
- **Some Related Method Developments**

1. Consistent Analysis of Independent Trajectories

Density matrix: $\rho_{ij}^{\text{cc}}(R,t) = \left| \rho_{ij}^{\text{cc}}(R,t) \right| \exp\left(i\theta_{ij}^{\text{cc}}(R,t)\right) = \left(\rho_{ii}^{\text{as}}(R,t) \rho_{jj}^{\text{as}}(R,t) \right)^{1/2} \frac{\rho_{ij}^{\text{wf}}(R,t)}{\left| \rho_{ij}^{\text{wf}}(R,t) \right|}$

Reduced density matrix: $\sigma_{ij}^{\text{cc}}(t) = \int \rho_{ij}^{\text{cc}}(R,t) dR$



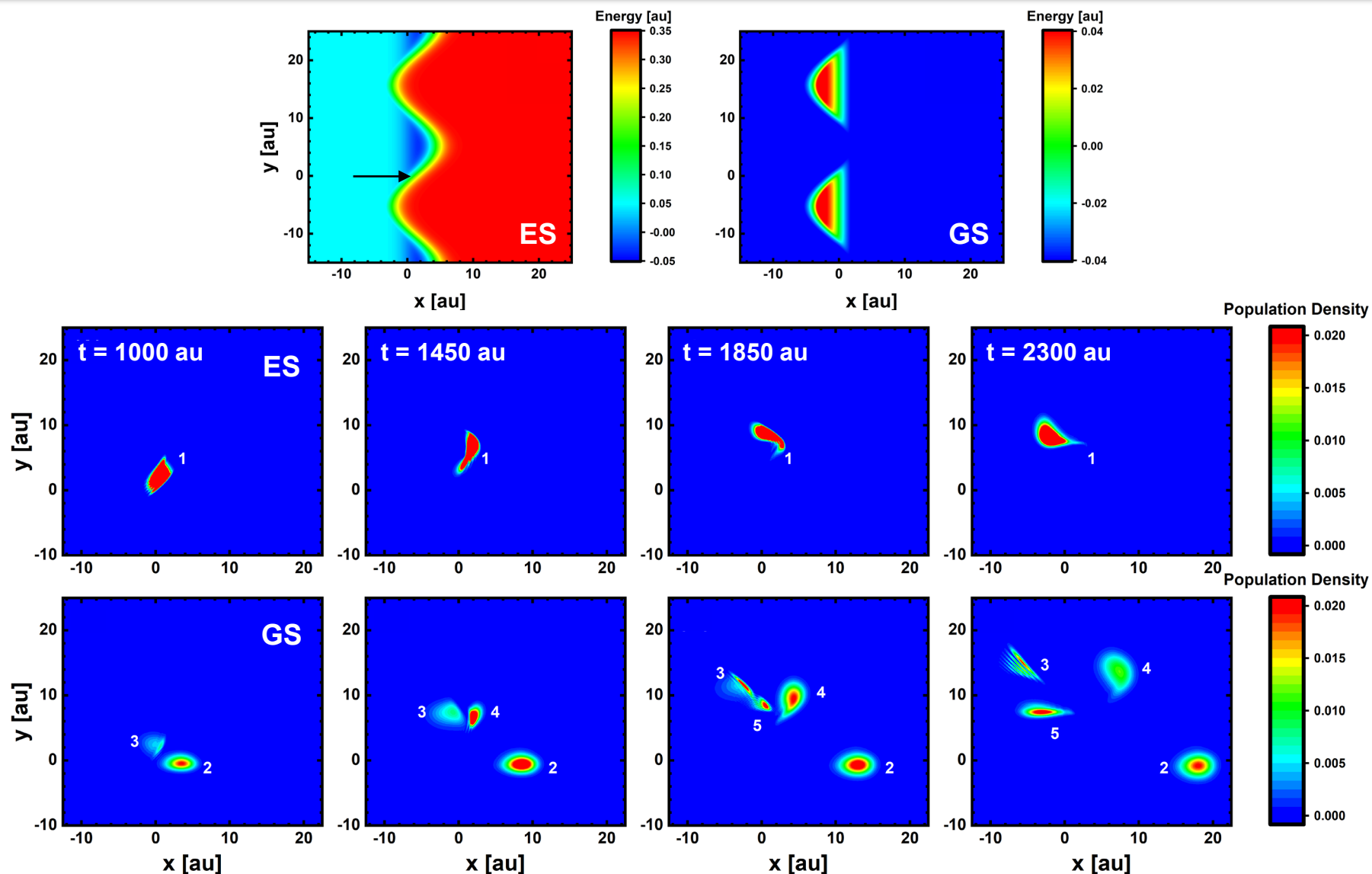
Landry, Falk, and Subotnik, *J. Chem. Phys.* 139, 211101 (2024)

Agostini, Min, Abedi, and Gross, *J. Chem. Theory Comput.* 12, 2127 (2016)

L. Wang* et al., *J. Chem. Theory Comput.* 20, 2349 (2024)

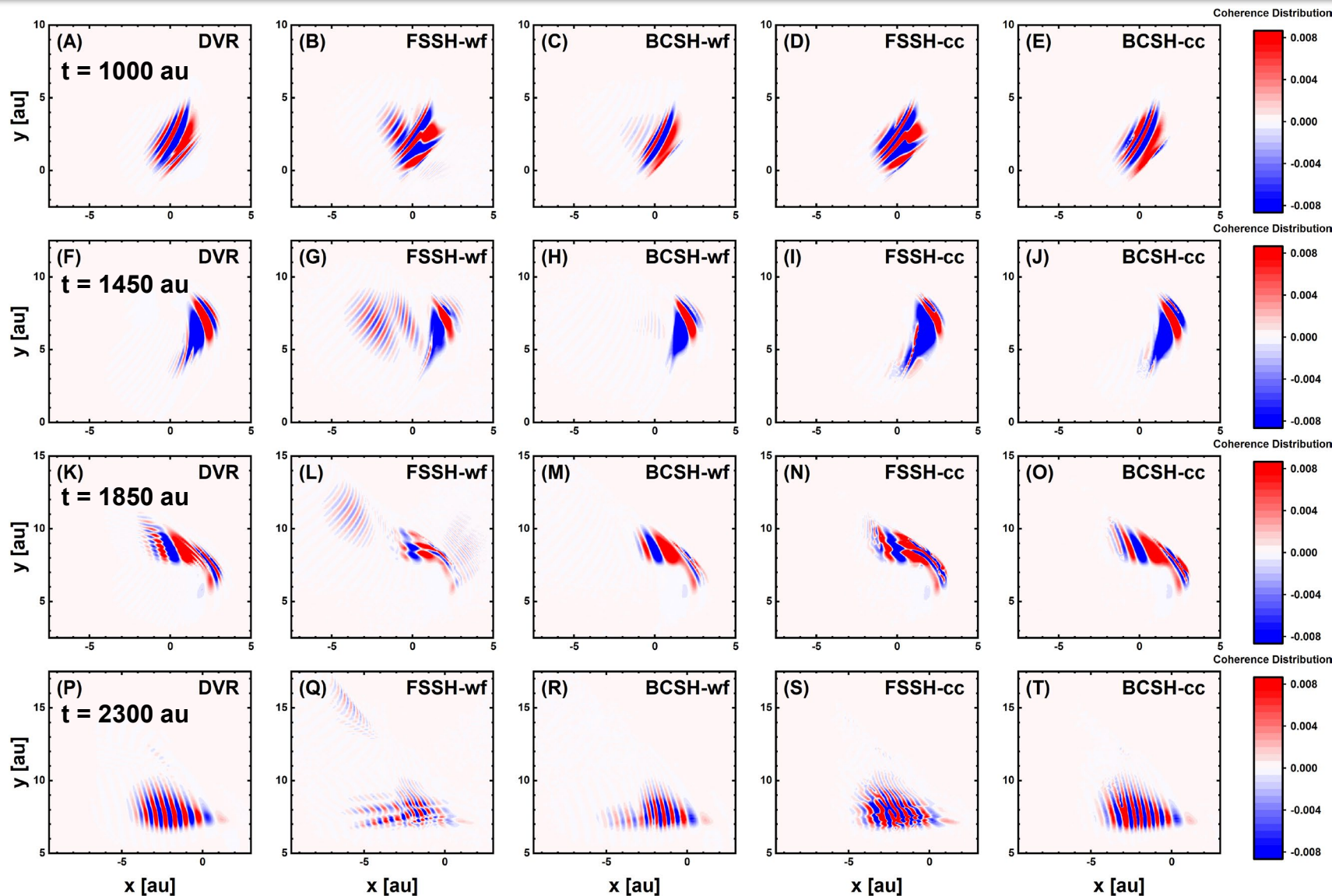
L. Wang* et al., *J. Chem. Phys.* 150, 164101 (2019)

Time Dependence and Spatial Distribution of Density Matrix



Subotnik, *J. Phys. Chem. A* 115, 12083 (2011)

Time Dependence and Spatial Distribution of Density Matrix



L. Wang* et al., *J. Chem. Theory Comput.* 20, 2349 (2024)

3. Sign-Consistent Coupled-Trajectory SH (SC-CTSH)

$$\Psi(\underline{\mathbf{R}}, \underline{\mathbf{r}}, t) = \chi(\underline{\mathbf{R}}, t) \Phi_{\underline{\mathbf{R}}}(\underline{\mathbf{r}}, t) \quad \text{Exact factorization}$$



$$i\hbar \frac{\partial}{\partial t} \chi(\underline{\mathbf{R}}, t) = \left[\sum_v \frac{(-i\hbar \nabla_v + \mathbf{A}_v(\underline{\mathbf{R}}, t))^2}{2M_v} + \epsilon(\underline{\mathbf{R}}, t) \right] \chi(\underline{\mathbf{R}}, t)$$

$$i\hbar \frac{\partial}{\partial t} \Phi_{\underline{\mathbf{R}}}(\underline{\mathbf{r}}, t) = [\hat{H}_{BO}(\underline{\mathbf{r}}, \underline{\mathbf{R}}) + \hat{U}_{en} - \epsilon(\underline{\mathbf{R}}, t)] \Phi_{\underline{\mathbf{R}}}(\underline{\mathbf{r}}, t)$$

$$\mathbf{A}_v(\underline{\mathbf{R}}, t) = \langle \Phi_{\underline{\mathbf{R}}}(\underline{\mathbf{r}}, t) | -i\hbar \nabla_v \Phi_{\underline{\mathbf{R}}}(\underline{\mathbf{r}}, t) \rangle_{\underline{\mathbf{r}}}$$

$$\epsilon(\underline{\mathbf{R}}, t) = \langle \Phi_{\underline{\mathbf{R}}}(\underline{\mathbf{r}}, t) | \hat{H}_{BO}(\underline{\mathbf{r}}, \underline{\mathbf{R}}) + \hat{U}_{en} - i\hbar \partial_t | \Phi_{\underline{\mathbf{R}}}(\underline{\mathbf{r}}, t) \rangle_{\underline{\mathbf{r}}}$$

$$\hat{U}_{en} = \sum_v \frac{1}{M_v} \left[\frac{(-i\hbar \nabla_v - \mathbf{A}_v(\underline{\mathbf{R}}, t))^2}{2} + \left(\frac{-i\hbar \nabla_v \chi(\underline{\mathbf{R}}, t)}{\chi(\underline{\mathbf{R}}, t)} + \mathbf{A}_v(\underline{\mathbf{R}}, t) \right) (-i\hbar \nabla_v - \mathbf{A}_v(\underline{\mathbf{R}}, t)) \right]$$

$$\text{CTSH:} \quad \dot{\mathbf{p}}_v^{(I)}(t) = -\nabla_v \epsilon_a^{(I)}$$

$$\text{SC-CTSH:}$$

$$\Phi_{\underline{\mathbf{R}}}(\underline{\mathbf{r}}, t) = \sum_l C_l(\underline{\mathbf{R}}, t) \phi_{\underline{\mathbf{R}}}^l(\underline{\mathbf{r}}, t)$$



$$\text{CT-MQC}$$

$$\dot{C}_l^{(I)}(t) = \dot{C}_{l,\text{Eh}}^{(I)}(t) + \dot{C}_{l,\text{XF}}^{(I)}(t) \quad \dot{\mathbf{p}}_v^{(I)}(t) = \dot{\mathbf{p}}_{v,\text{Eh}}^{(I)}(t) + \dot{\mathbf{p}}_{v,\text{XF}}^{(I)}(t)$$

$$\dot{C}_{l,\text{Eh}}^{(I)}(t) = -\frac{i}{\hbar} \epsilon_l^{(I)} C_l^{(I)} - \sum_k \sum_{v=1}^{N_n} \frac{\mathbf{p}_v^{(I)}(t)}{M_v} \cdot \mathbf{d}_{v,lk}^{(I)} C_k^{(I)}$$

$$\dot{\mathbf{p}}_{v,\text{Eh}}^{(I)}(t) = -\sum_l \rho_{ll}^{(I)}(t) \nabla_v \epsilon_l^{(I)} - \sum_{lk} \rho_{lk}^{(I)}(t) (\epsilon_k^{(I)} - \epsilon_l^{(I)}) \mathbf{d}_{v,lk}^{(I)}$$

$$\dot{C}_{l,\text{XF}}^{(I)}(t) = -\sum_m \rho_{mm}^{(I)} \left[\sum_v \frac{\mathbf{Q}_v^{(I)}(t)}{\hbar M_v} \cdot (\mathbf{f}_{v,m}^{(I)}(t) - \mathbf{f}_{v,l}^{(I)}(t)) \right] C_l^{(I)}$$

$$\dot{\mathbf{p}}_{v,\text{XF}}^{(I)}(t) = -\sum_{l,m} \rho_{ll}^{(I)} \rho_{mm}^{(I)} \left[\sum_v \frac{2\mathbf{Q}_v^{(I)}(t)}{\hbar M_v} \cdot (\mathbf{f}_{v,m}^{(I)}(t) - \mathbf{f}_{v,l}^{(I)}(t)) \right] \mathbf{f}_{v,l}^{(I)}(t)$$

Quantum momentum

$$\mathbf{Q}_v^{(I)}(t) = -\hbar \left(\nabla_v |\chi^{(I)}(t)|^2 \right) / \left(2 |\chi^{(I)}(t)|^2 \right)$$

Sign consistency

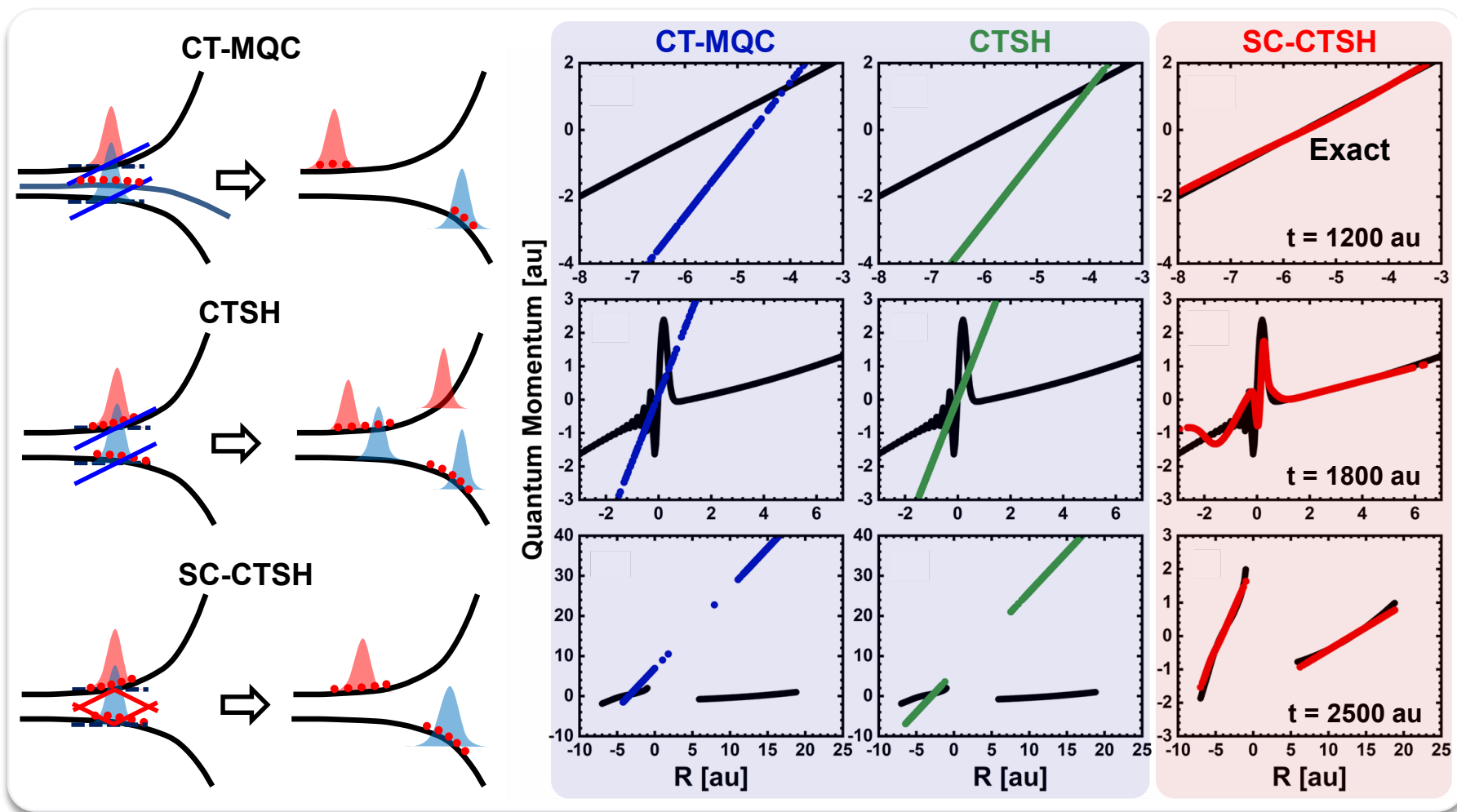
$$\dot{C}_{l,\text{XF,SC}}^{(I)}(t) = -\sum_m \rho_{mm}^{(I)} \left[\sum_v \frac{\mathbf{Q}_v^{(I)}(t)}{\hbar M_v} \cdot (\mathbf{f}_{v,m}^{(I)}(t) - \mathbf{f}_{v,l}^{(I)}(t)) \right] C_l^{(I)} \cdot \lambda^{(I)}(t)$$

CT-MQC: Min, Agostini, and Gross, *Phys. Rev. Lett.* 115, 073001 (2015)

CTSH: Pieroni and Agostini, *J. Chem. Theory Comput.* 17, 5969 (2021)

SC-CTSH: L. Wang* et al., *J. Chem. Phys.* 162, 164103 (2025)

Quantum Momentum by SC-CTSH

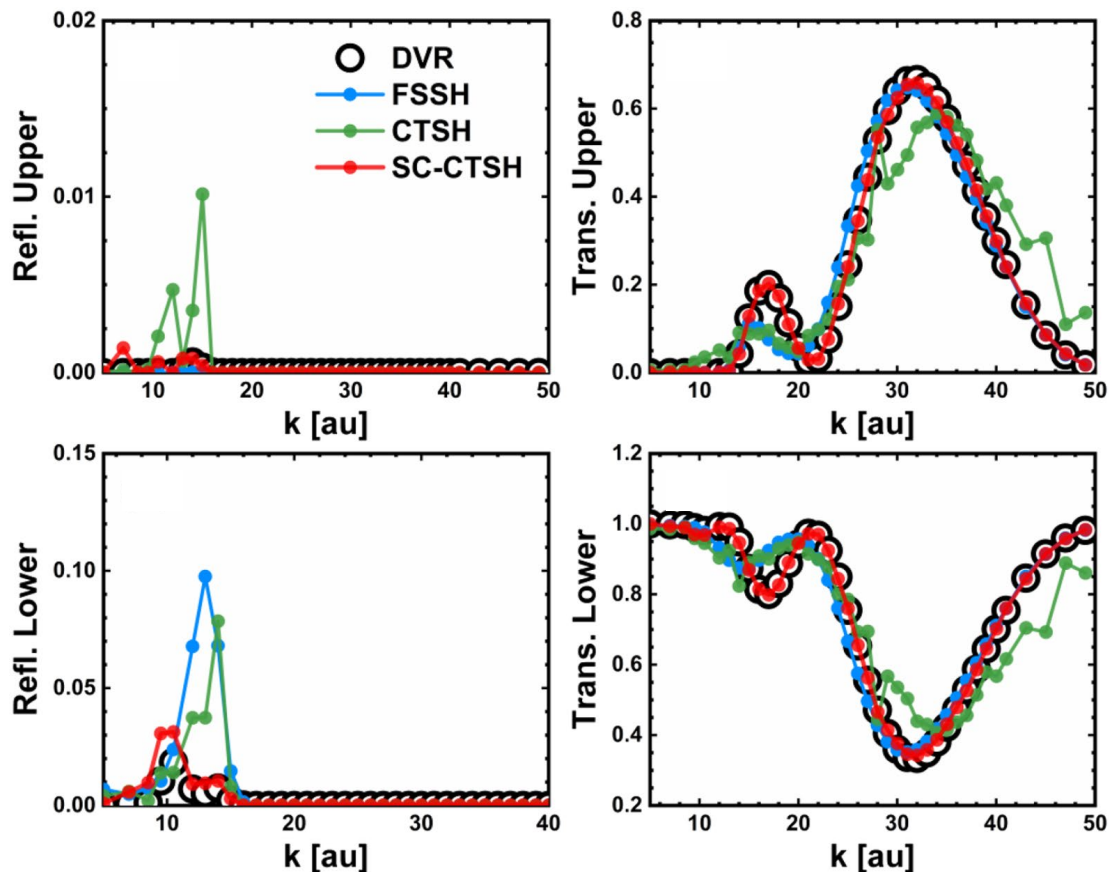


SC-CTSH gives the correct quantum momentum and decoherence

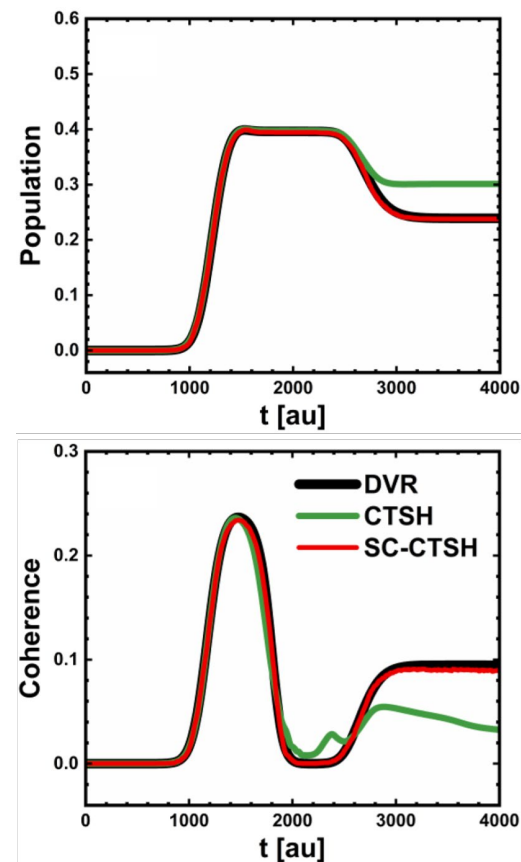
L. Wang* et al., *J. Chem. Phys.* 162, 164103 (2025)

Performance of SC-CTSH

DAC model



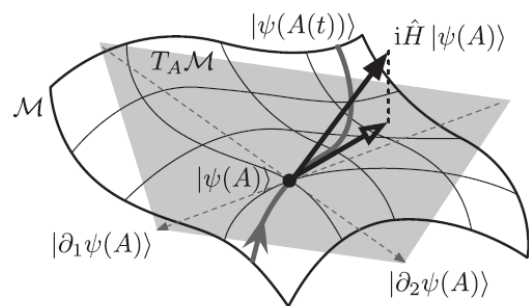
ECR model



SC-CTSH gives correct channel populations and coherence

L. Wang* et al., *J. Chem. Phys.* 162, 164103 (2025)

4. Time-Dependent Variation with Trajectory Basis



- ✓ Nuclear tunneling
- ✓ Zero-point energy
- ✓ Quantum interference

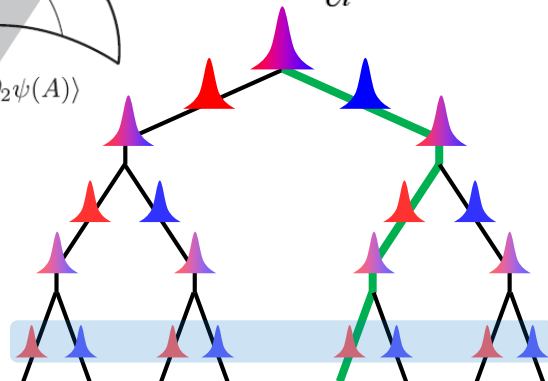
$$\langle \delta \Psi | H - i\hbar \frac{\partial}{\partial t} | \Psi \rangle = 0$$

Configuration:

$$|\psi(t)\rangle = |\phi_{\text{system}}(t)\rangle |\mathbf{z}_{\text{bath}}(t)\rangle$$

$$|\mathbf{z}_{\text{bath}}(t)\rangle = \prod_{m=1}^M |z^m(t)\rangle$$

$$\langle x | z^m \rangle = \left(\frac{\gamma}{\pi} \right)^{1/4} \exp \left[-\frac{\gamma}{2} (x - q^m)^2 + i \frac{p^m (x - q^m)}{\hbar} + \frac{i p^m q^m}{2\hbar} \right]$$



Initial condition:

$$\hat{\rho}_{\text{tot}}(0) = \hat{\rho}_b^{\text{eq}} \otimes \hat{\rho}_s(0)$$

$$\begin{aligned} \hat{\rho}_b^{\text{eq}} &= \int \frac{d^2 \mathbf{z}_{\text{init}}}{\pi^M} |\mathbf{z}_{\text{init}}\rangle \rho_b(\mathbf{z}_{\text{init}}) \langle \mathbf{z}_{\text{init}}| \\ &= \prod_{m=1}^M \int d^2 z_{\text{init}}^m |z_{\text{init}}^m\rangle \rho_b(z_{\text{init}}^m) \langle z_{\text{init}}^m| \end{aligned}$$

$$\rho_b(z_{\text{init}}^m) = \frac{e^{\beta \omega^m} - 1}{\pi} e^{-[(\text{Re}(z_{\text{init}}^m) + \zeta_x)^2 + (\text{Im}(z_{\text{init}}^m) + \zeta_y)^2]} (e^{\beta \omega^m} - 1)$$

$$|\Psi_0(0)\rangle = |\mathbf{z}_{\text{init}}\rangle (|1\rangle + 0|2\rangle)$$



$$|\Psi(t=0)\rangle = \sum_{j=1}^N |\mathbf{z}_j\rangle (a_j^{(1)} |1\rangle + a_j^{(2)} |2\rangle)$$

MCEv1:

$$|\Psi(t)\rangle = \sum_{j=1}^N |\psi_j(t)\rangle = \sum_{j=1}^N \sum_s a_j^{(s)}(t) |\mathbf{z}_j(t)\rangle |s\rangle \Rightarrow \sum_{j=1}^N i\hbar \langle \mathbf{z}_l | \mathbf{z}_j \rangle \dot{a}_j^{(s)} = \sum_{j=1}^N \sum_{s'} \langle \mathbf{z}_l | \hat{H}_{ss'} | \mathbf{z}_j \rangle a_j^{(s')} - \sum_{j=1}^N i\hbar \langle \mathbf{z}_l | \dot{\mathbf{z}}_j \rangle a_j^{(s)}$$

A proper trajectory basis is important for time-dependent variation

Martínez, Ben-Nun, and Levine, *J. Phys. Chem.* 100, 7884 (1996)

Shalashilin, *J. Chem. Phys.* 130, 244101 (2009)

Multiconfigurational Surface Hopping (MCSH)

$$|\Psi(t)\rangle = \sum_{j=1}^N |\psi_j(t)\rangle = \sum_{j=1}^N \sum_s a_j^{(s)}(t) |\mathbf{z}_j(t)\rangle |s\rangle$$

$$\langle \partial \Psi | H - i\hbar \frac{\partial}{\partial t} | \Psi \rangle = 0$$

$$\sum_{j=1}^N i\hbar \langle \mathbf{z}_i | \mathbf{z}_j \rangle \dot{a}_j^{(s)} = \sum_{j=1}^N \sum_{s'} \langle \mathbf{z}_i | \hat{H}_{ss'} | \mathbf{z}_j \rangle a_j^{(s')} - \sum_{j=1}^N i\hbar \langle \mathbf{z}_i | \dot{\mathbf{z}}_j \rangle a_j^{(s)}$$

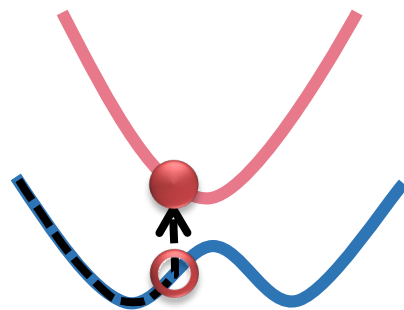
$$\dot{\mathbf{p}}_{SH,j} = -\nabla_{\mathbf{q}_{SH,j}} E_a \quad \dot{\mathbf{q}}_{SH,j} = \mathbf{p}_{SH,j}$$

MQCD trajectories

$$\mathbf{z}_j = \left(\frac{\gamma}{2}\right)^{\frac{1}{2}} \mathbf{q}_{SH,j} + \frac{i}{\hbar} \left(\frac{1}{2\gamma}\right)^{\frac{1}{2}} \mathbf{p}_{SH,j}$$

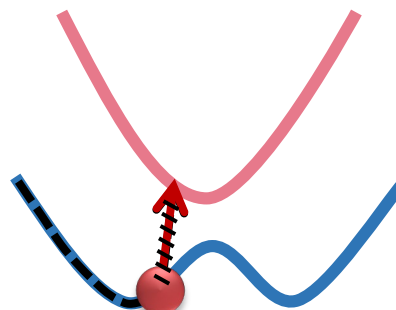
$$\dot{\mathbf{z}}_j = \left(\frac{\gamma}{2}\right)^{\frac{1}{2}} \dot{\mathbf{q}}_{SH,j} + \frac{i}{\hbar} \left(\frac{1}{2\gamma}\right)^{\frac{1}{2}} \dot{\mathbf{p}}_{SH,j}$$

Surface hopping trajectories cannot be directly used because momenta are discontinuous



Surface hop
at time t

Return to
time $t - dt$



Linear interpolation
during $[t - dt, t]$

$$\dot{\mathbf{p}}_{interpolation} = \frac{\mathbf{p}(t) - \mathbf{p}(t - dt)}{dt}$$

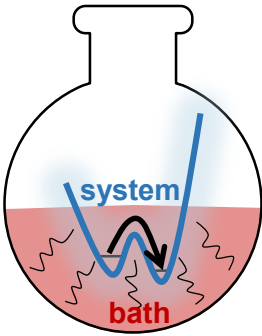
$$\dot{\mathbf{q}}_{interpolation} = \frac{\mathbf{q}(t) - \mathbf{q}(t - dt)}{dt}$$

Linear interpolation generates continuous trajectories and can be used for time variation

Time-dependent variation with momentum-jump trajectories

L. Wang* et al., *J. Chem. Theory Comput.* 20, 7755 (2024)

Spin-Boson Models at Low Temperatures



Hamiltonian

$$\hat{H}_s = \varepsilon \hat{\sigma}_z + \Delta \hat{\sigma}_x \quad \hat{H}_b = \sum_m^M \omega^m (\hat{b}^{m\dagger} \hat{b}^m + \frac{1}{2})$$

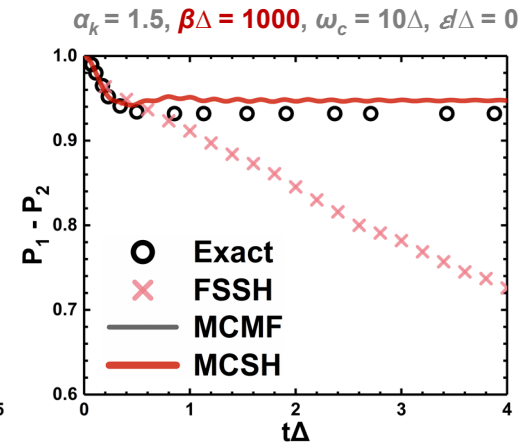
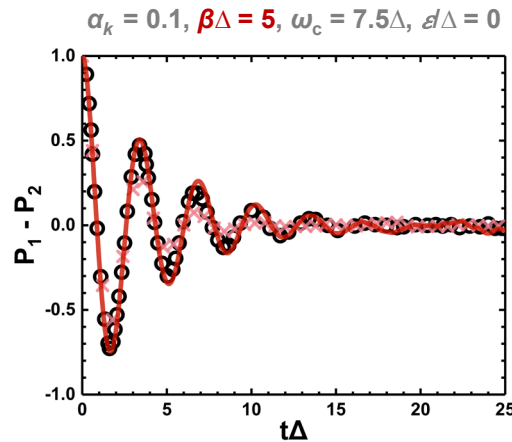
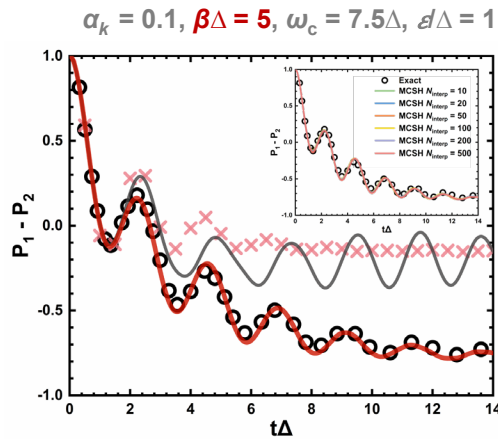
$$\hat{H}_{sb} = \hat{\sigma}_z \sum_m^M \frac{c^m}{\sqrt{2\omega^m}} (\hat{b}^{m\dagger} + \hat{b}^m)$$

Ohmic spectral density

$$J(\omega) = \frac{\pi}{2} \alpha_k \omega e^{-\omega/\omega_c}$$

Initial density matrix

$$\hat{\rho}_{tot}(0) = |1\rangle\langle 1| \otimes \hat{\rho}_b^{eq}$$



MCSH consistently shows high performance and makes significant improvements compared with FSSH

Summary

Quantum Dynamics
Time-dependent variation with trajectory basis MCSH, MCE-HEOM
Branching with coupled trajectories DCC, SC-CTSH
Branching with auxiliary trajectories A-BCMF, A-BCSH
Branching with independent trajectories BCMF, BCSH
Time-dependent self- consistent field approximation EMF, FSSH
Classical path approximation CPA

- The **principle of detailed internal consistency** has been proposed as the fundamental principle (or basic relation) for mixed quantum-classical dynamics, which is purely due to **logical consistency** $\rho_{ii}^{\text{wf}}(R, t) = \rho_{ii}^{\text{as}}(R, t)$
- **A series of methods** have been proposed based on the detailed internal consistency, ranging from independent-trajectory, auxiliary-trajectory and coupled-trajectory frameworks
- **MCSH** can efficiently realize multiconfigurational variational dynamics based on different kinds of surface hopping-like trajectories
- **DCC** achieves accurate population, coherence and detailed balance, and is promising to be combined with MCSH to study complex systems

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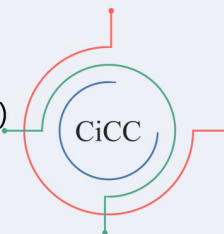
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